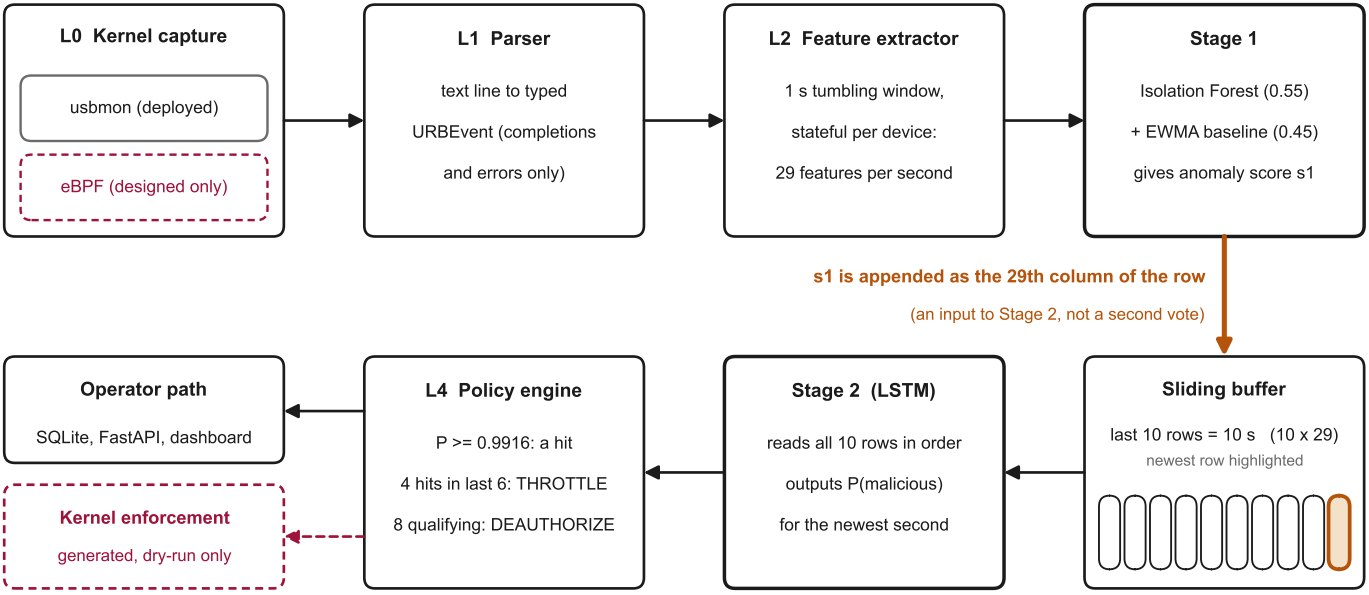
L0 Kernel capture
usbmon (deployed)
eBPF (designed only)
L1 Parser
text line to typed
URBEvent (completions
and errors only)
L2 Feature extractor
1 s tumbling window,
stateful per device:
29 features per second
Stage 1
Isolation Forest (0.55)
+ EWMA baseline (0.45)
gives anomaly score s1
s1 is appended as the 29th column of the row
(an input to Stage 2, not a second vote)
Sliding buffer
last 10 rows = 10 s (10 x 29)
newest row highlighted
Stage 2 (LSTM)
reads all 10 rows in order
outputs P(malicious)
for the newest second
L4 Policy engine
P >= 0.9916: a hit
4 hits in last 6: THROTTLE
8 qualifying: DEAUTHORIZE
Operator path
SQLite, FastAPI, dashboard
Kernel enforcement
generated, dry-run only